# Monitoring the System Performance with Prometheus and Grafana

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**Operating System: macOS with Apple M4 Chip**

This project demonstrates how to monitor system performance using Prometheus and Grafana on a macOS device running an Apple M4 chip. The setup enables real-time collection and visualization of system metrics such as CPU usage, memory, and disk activity.

## Prometheus Configuration

Prometheus is an open-source monitoring and alerting toolkit. It scrapes metrics from instrumented targets at given intervals, evaluates rule expressions, and can trigger alerts if conditions are met. In this project, Prometheus scrapes system metrics from the Node Exporter.

## Node Exporter Setup

Node Exporter is a Prometheus exporter for hardware and OS metrics exposed by *nix kernels. It runs as a separate service and exposes metrics on port 9100. The metrics include CPU load, memory usage, disk I/O, and more.

## Grafana Dashboard

Grafana is an open-source analytics and monitoring solution. It integrates with Prometheus to visualize the collected metrics. In this project, the Node Exporter Full dashboard was used to display system stats. The Prometheus data source was added manually under Grafana's Connections > Data Sources section.

## Important Commands and Tips

To restart Prometheus:

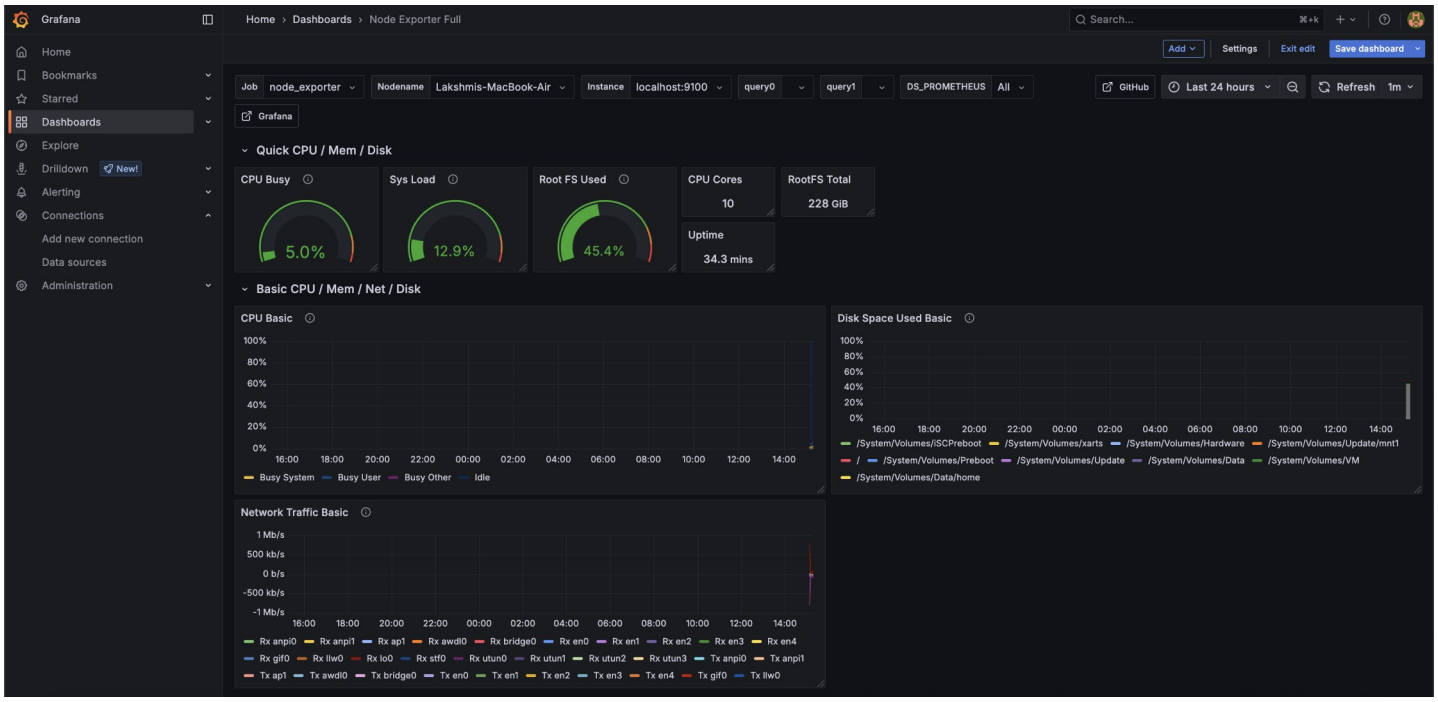
```
/opt/homebrew/bin/prometheus --config.file=prometheus.yml
```

To restart Node Exporter:

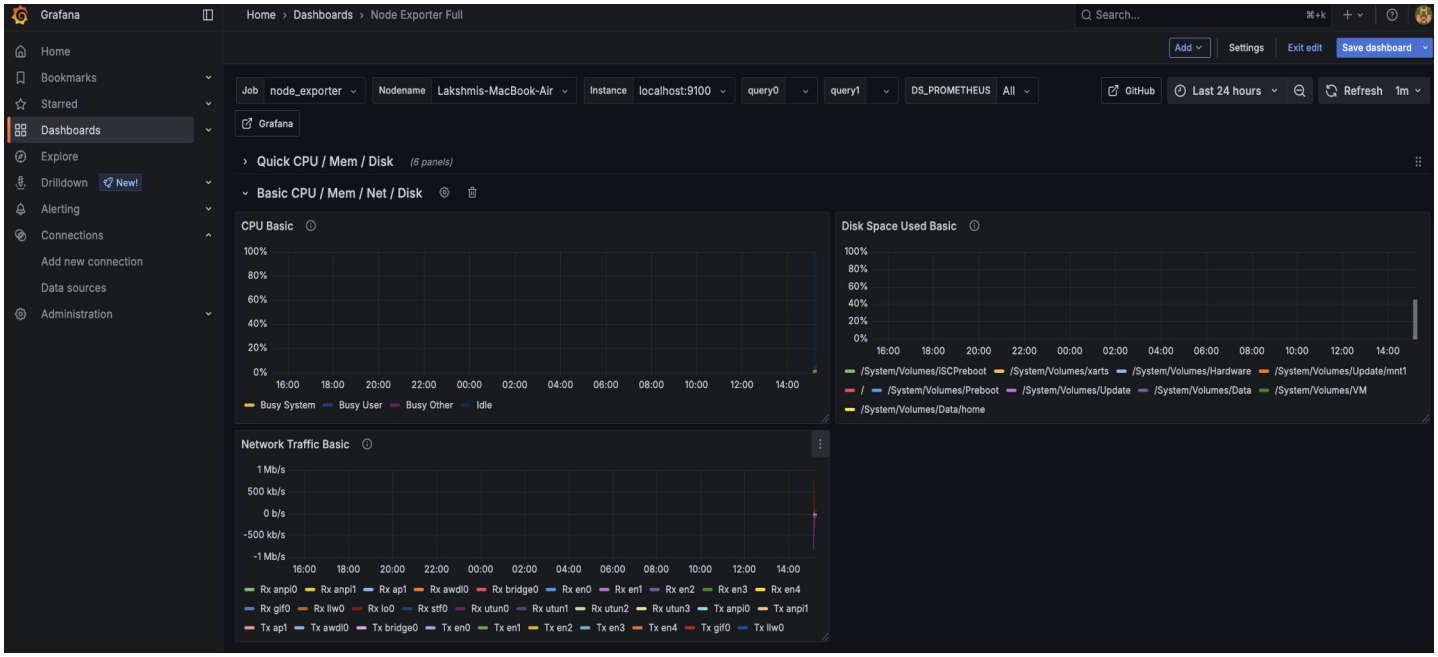
```
./node_exporter
```

In Grafana, if panels show 'No data', go to Dashboard Settings > Variables and ensure the DS_PROMETHEUS variable is linked to the correct running Prometheus instance (e.g., 'prometheus-1'). Then save and refresh the dashboard.

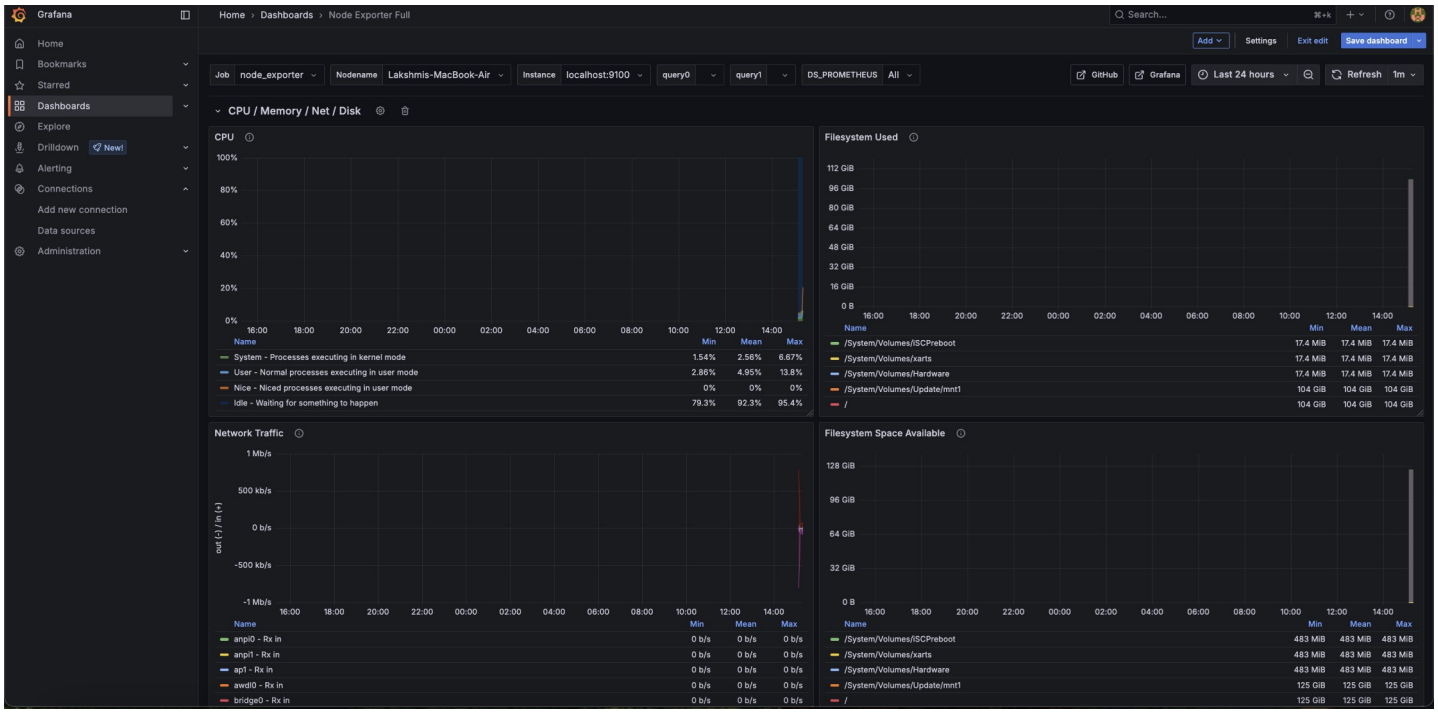
# Quick CPU/Mem/Disk

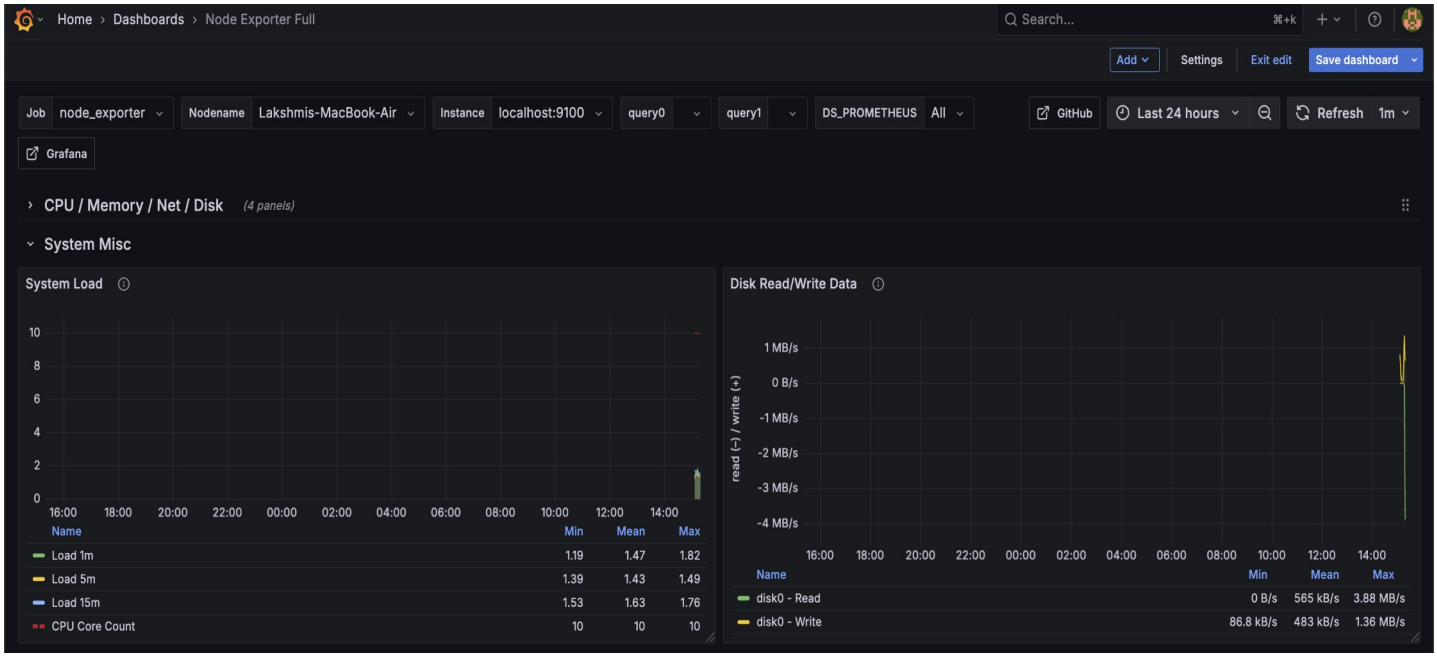



# Basic CPU/Mem/Net/Disk

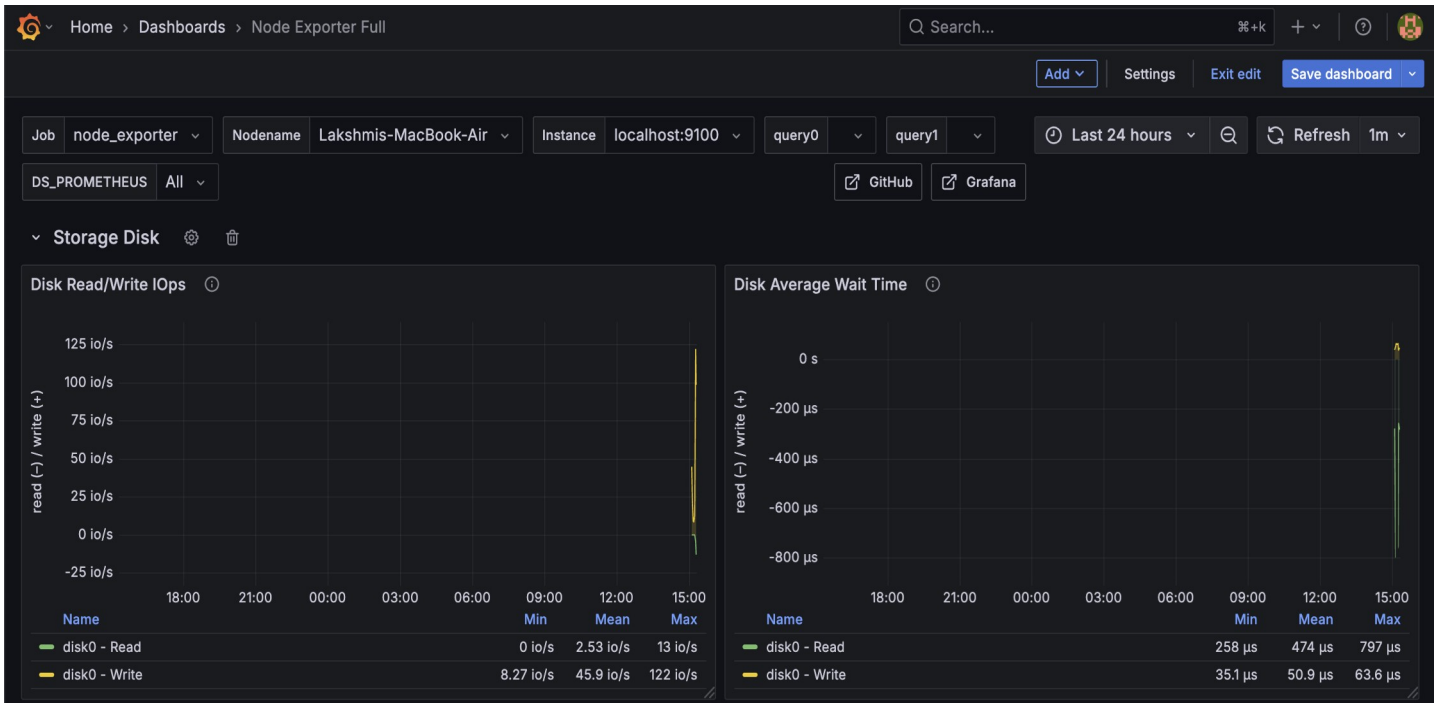


# CPU/Memory/Net/Disk

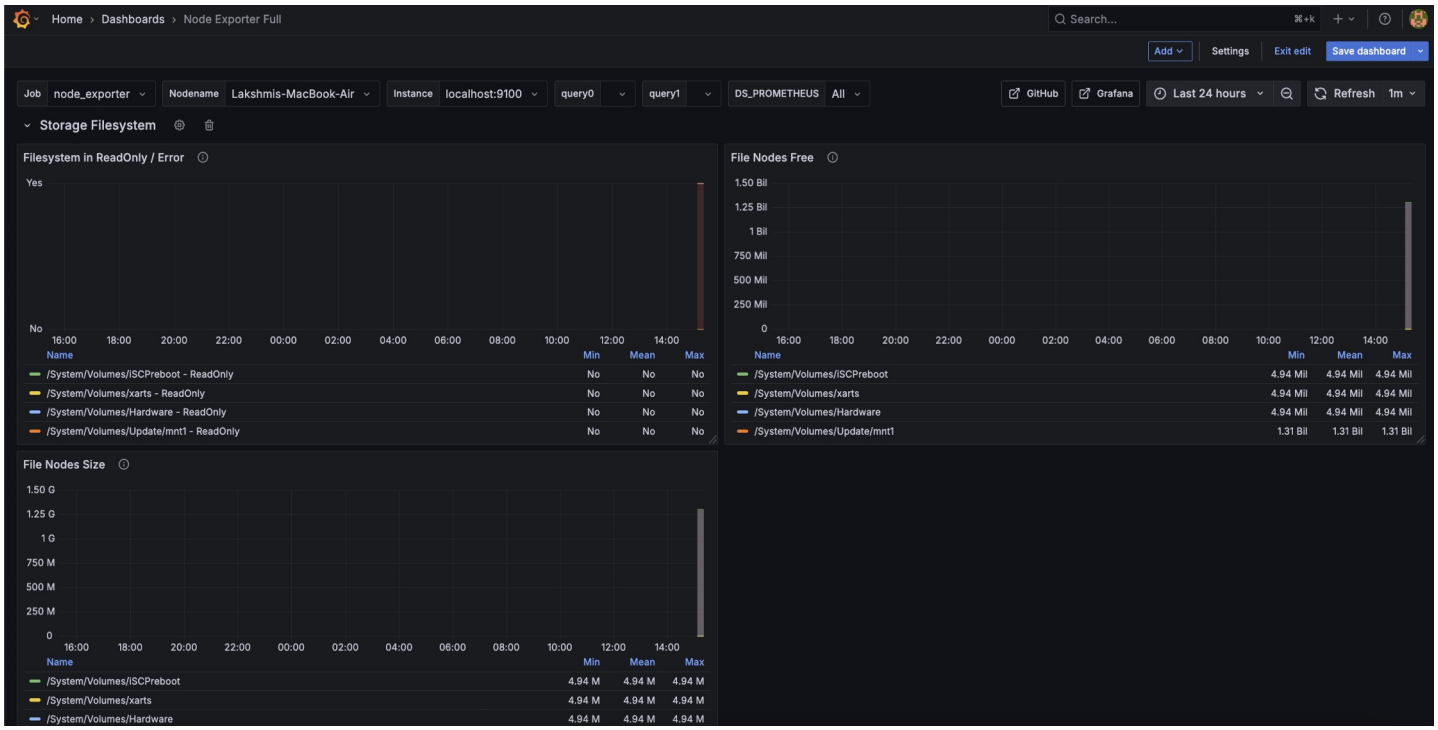


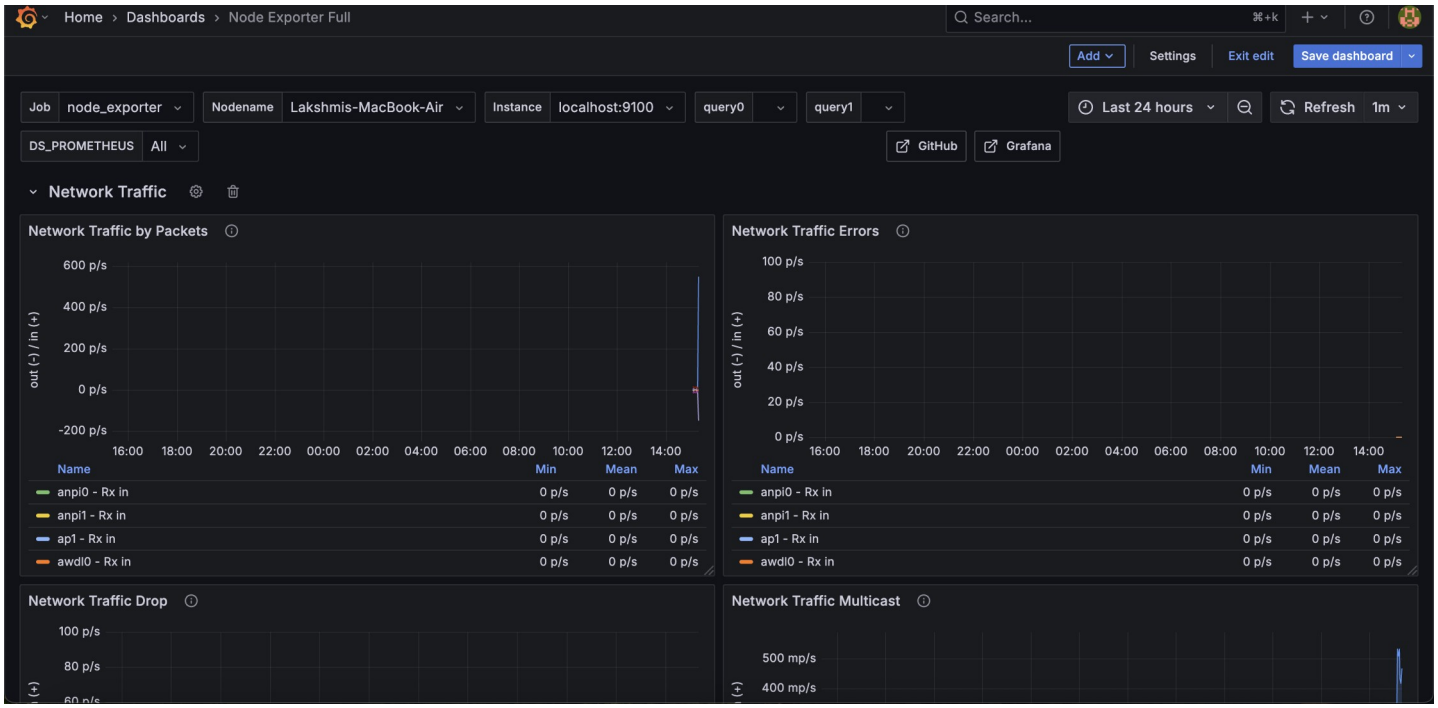
# STORAGE DISK

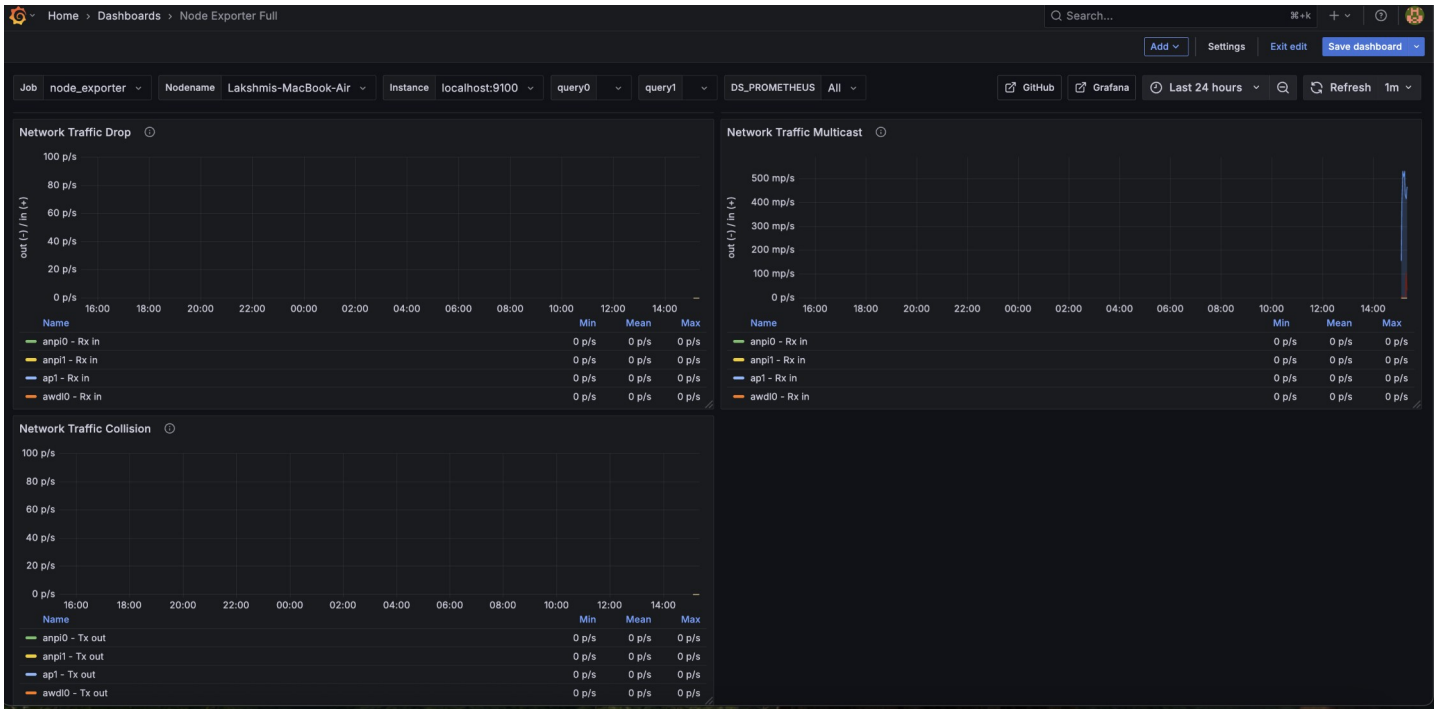
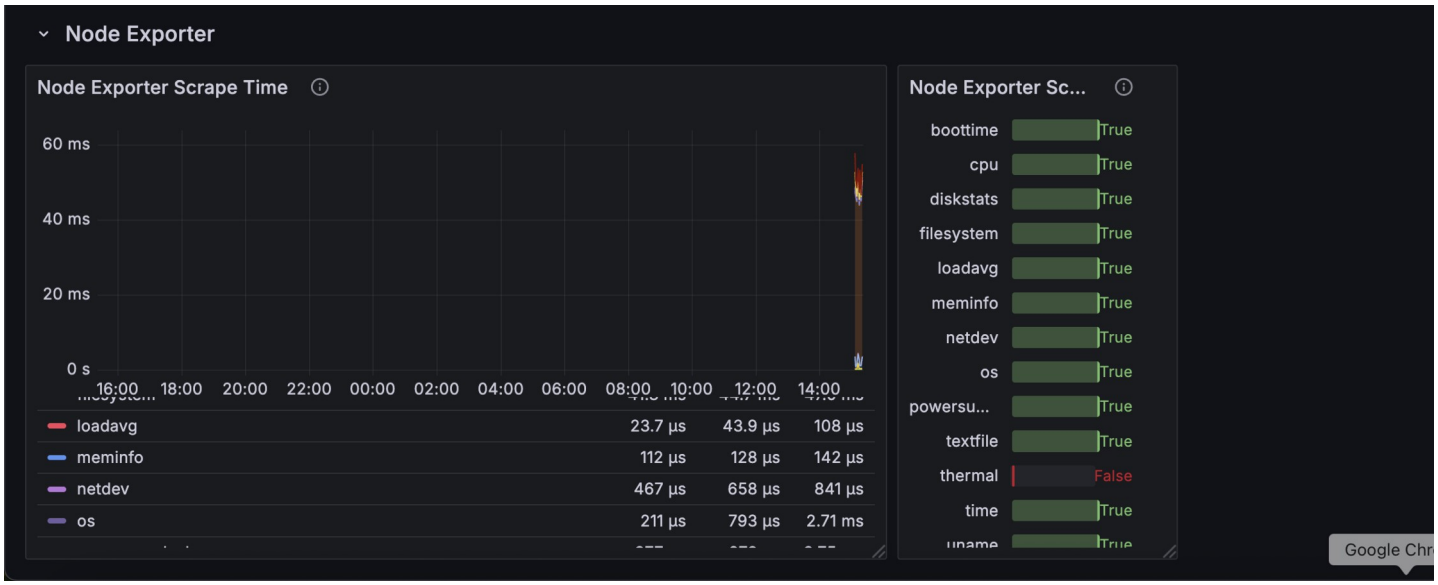



# STORAGE FILE SYSTEM



# NETWORK TRAFFIC






# NODE EXPORTER